

Spectral Bands

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```
library(dplyr)
library(tidyr)
library(tidyverse)
library(ggplot2)
library(corrplot)
library(psych)
library(readxl)

cn_multiple <- read_excel("CN_2_table.xlsx", sheet = "JuneN")
cn_single <- read_excel("CN_2_table.xlsx", sheet = "RawN")

y <- "Total_N"
single_band <- colnames(cn_single)[2:5]
mult_band <- colnames(cn_multiple)[2:7]

spectral_cols <- c(single_band, mult_band)

cn_single_mult <- cbind(cn_multiple, cn_single)

X <- cn_single_mult %>% dplyr::select(all_of(spectral_cols))

X_scaled <- scale(X, center = T, scale = T)

PCA:

pca_single <- prcomp(X %>% dplyr::select(all_of(single_band)), center = T, scale. = T)
pca_mult <- prcomp(X %>% dplyr::select(all_of(mult_band)), center = T, scale. = T)

# View summary
summary(pca_mult)

## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation  2.2054 1.0657 0.02188 0.006771 0.0002158 7.375e-20
## Proportion of Variance 0.8106 0.1893 0.00008 0.000010 0.0000000 0.000e+00
## Cumulative Proportion 0.8106 0.9999 0.99999 1.000000 1.0000000 1.000e+00

# Standard deviations of PCs
pca_single$sdev

## [1] 1.6571993 0.9652725 0.4697274 0.3182699

# Loadings (eigenvector coefficients)
pca_mult$rotation
```

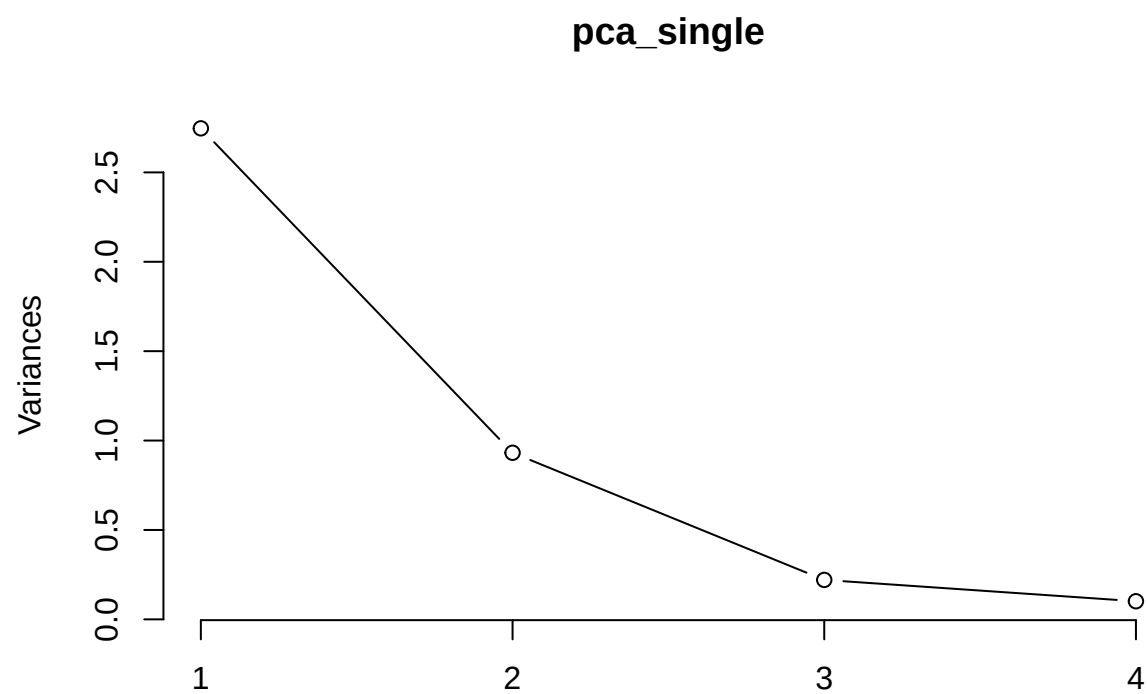
```
##              PC1              PC2              PC3              PC4
## Green_NDVI_June2_Clip -0.1585582  0.87909273  0.22242481  0.3905958
## SR_June2_Clip -0.4532371  0.02320451 -0.73151188  0.1752453
## MTVI2_June2_Clip -0.3913261 -0.47392359  0.39567664  0.6823832
## MSAVI_June2_Clip -0.4532098  0.02752368  0.49252708 -0.5319237
## NDVI_June2_Clip -0.4532660  0.02545987 -0.09020713 -0.1845728
## Red_Edge_NDVI_June2_Clip -0.4532660  0.02545987 -0.09020713 -0.1845728
##              PC5              PC6
## Green_NDVI_June2_Clip -0.004171763  0.000000e+00
## SR_June2_Clip -0.477720758  4.120263e-17
## MTVI2_June2_Clip -0.007308334  7.289658e-19
## MSAVI_June2_Clip -0.517993747  4.063864e-17
## NDVI_June2_Clip  0.501694399  7.071068e-01
## Red_Edge_NDVI_June2_Clip  0.501694399 -7.071068e-01
```

```
# PC scores for observations
head(pca_single$x)
```

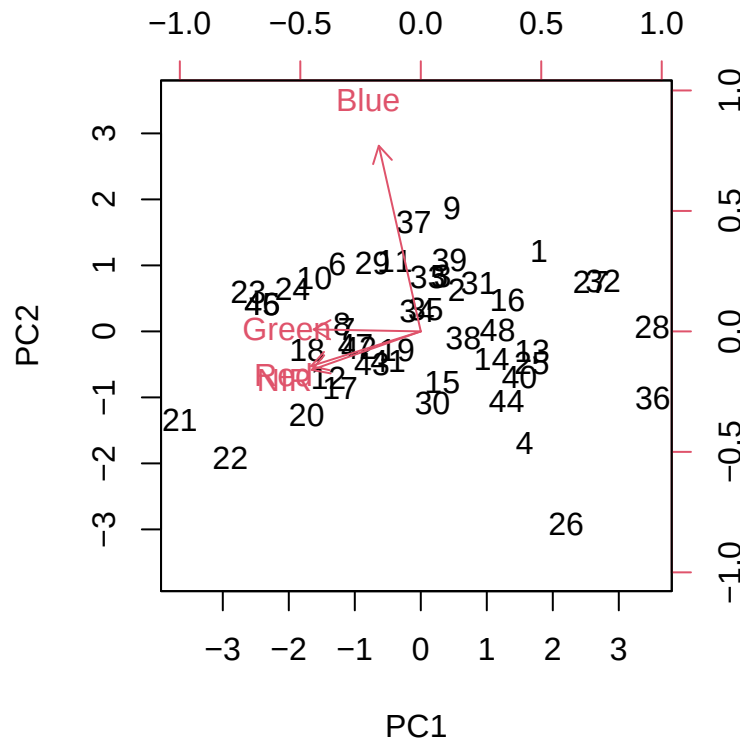
```
##              PC1              PC2              PC3              PC4
## [1,]  1.7913361  1.2167778  0.60150084  0.08496092
## [2,]  0.5449352  0.6385740  0.05481045 -0.12954254
## [3,]  0.3305895  0.8333113  0.76574144  0.50367286
## [4,]  1.5743094 -1.6887891 -0.31509030 -0.46849296
## [5,]  0.2833373  0.8399538 -0.29698281  0.16672343
## [6,] -1.2634449  1.0242765 -0.37621114  0.60865000
```

Scree plot and bi plot for single:

```
plot(pca_single, type = "l")
```



```
biplot(pca_single, scale = 0)
```



```
k <- 2
pcs_single <- data.frame(pca_single$x[,1:k])
pcs_mult <- data.frame(pca_mult$x[,1:k])
colnames(pcs_single) <- c("PC1_Single", "PC2_Single")
colnames(pcs_mult) <- c("PC1_Mult", "PC2_Mult")

pca_df <- cbind(pcs_single, pcs_mult)
```

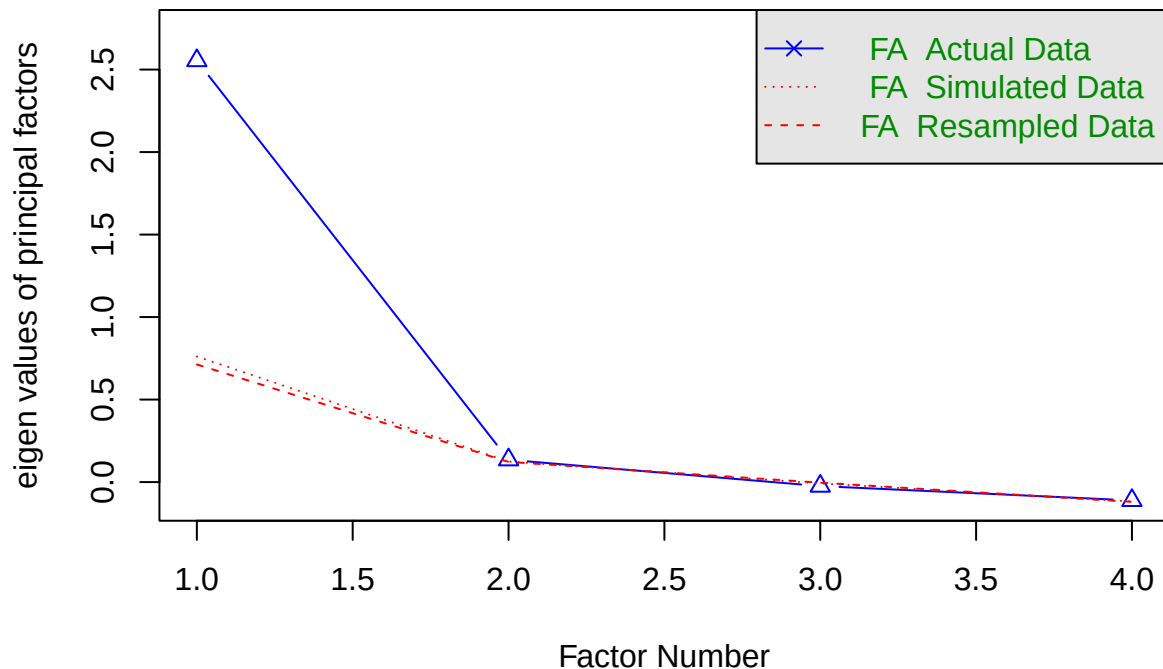
Factor Analysis:

```
parallel_model_single <- fa.parallel(X %>% dplyr::select(all_of(single_band)),
  fm = "pa",
  fa = "fa",
  n.iter = 50,
  main = "Parallel Analysis Scree Plot - Single Bands")
```

```
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
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## ultra-Heywood case was detected. Examine the results carefully
```

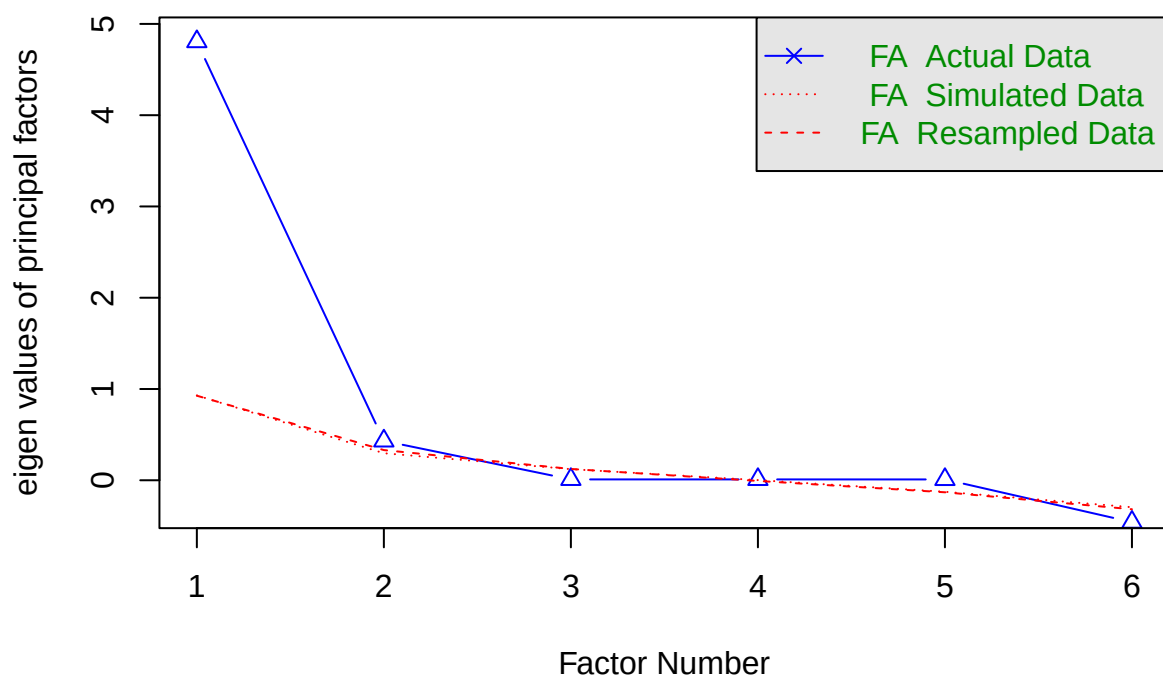
Parallel Analysis Scree Plot – Single Bands



```
## Parallel analysis suggests that the number of factors = 1 and the number of components = NA
parallel_model_mult <- fa.parallel(X %>% dplyr::select(all_of(mult_band)),
  fm = "pa",
  fa = "fa",
  n.iter = 50,
  main = "Parallel Analysis Scree Plot - Multiple Bands")
```

```
## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done
## Warning in log(det(m.inv.r)): NaNs produced
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## In factor.scores, the correlation matrix is singular, the pseudo inverse is used
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected. Examine the results carefully
```


Parallel Analysis Scree Plot – Multiple Bands



```
## Parallel analysis suggests that the number of factors = 2 and the number of components = NA
```

```
efa_model_single <- fa(r = X %>% dplyr::select(all_of(single_band)),
  nfactors = 2,
  fm = "pa",
  rotate = "oblimin",
  scores = "regression")
```

```
## Loading required namespace: GPArotation
```

```
efa_model_mult <- fa(r = X %>% dplyr::select(all_of(mult_band)),
  nfactors = 2,
  fm = "pa",
  rotate = "oblimin",
  scores = "regression")
```

```
## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done
```

```
## In factor.stats, I could not find the RMSEA upper bound . Sorry about that
```

```
## In factor.scores, the correlation matrix is singular, the pseudo inverse is used
```

```
Loadings:
```

```
print(efa_model_single$loadings)
```

```
##
```

```
## Loadings:
```

```
##      PA1    PA2
## Blue      0.470
```

```
## Green 0.668 0.376
## Red 0.957
## NIR 0.980
##
##          PA1    PA2
## SS loadings 2.322 0.367
## Proportion Var 0.581 0.092
## Cumulative Var 0.581 0.672
```

Loadings:

```
print(efa_model_mult$loadings)
```

```
##
## Loadings:
##          PA1    PA2
## Green_NDVI_June2_Clip 0.118 0.975
## SR_June2_Clip 0.976 0.113
## MTVI2_June2_Clip 0.970 -0.433
## MSAVI_June2_Clip 0.975 0.118
## NDVI_June2_Clip 0.976 0.116
## Red_Edge_NDVI_June2_Clip 0.976 0.116
##
##          PA1    PA2
## SS loadings 4.763 1.192
## Proportion Var 0.794 0.199
## Cumulative Var 0.794 0.992
```

Loadings:

```
print(efa_model_single$weights)
```

```
##          PA1    PA2
## Blue -0.001135395 0.23895535
## Green 0.175474868 0.90505797
## Red 0.412222665 -0.07877152
## NIR 0.439176350 -0.27166769
```

```
print(efa_model_mult$weights)
```

```
##          PA1    PA2
## Green_NDVI_June2_Clip 0.18588487 0.2058940
## SR_June2_Clip 0.18455079 -0.1634744
## MTVI2_June2_Clip 0.54207774 -1.4596344
## MSAVI_June2_Clip 0.04614228 0.8685237
## NDVI_June2_Clip 0.11640782 0.3607637
## Red_Edge_NDVI_June2_Clip 0.11640782 0.3607637
```

```
lf_single <- efa_model_single$scores[, 1:2]
lf_mult <- efa_model_mult$scores[, 1:2]
colnames(lf_single) <- c("LF_1_single", "LF_2_single")
colnames(lf_mult) <- c("LF_1_mult", "LF_2_mult")
lf_df <- cbind(lf_single, lf_mult)
lf_df$Y <- cn_multiple$Total_N
```

```
## Warning in lf_df$Y <- cn_multiple$Total_N: Coercing LHS to a list
```

Fit Regression Models: